

HW N3

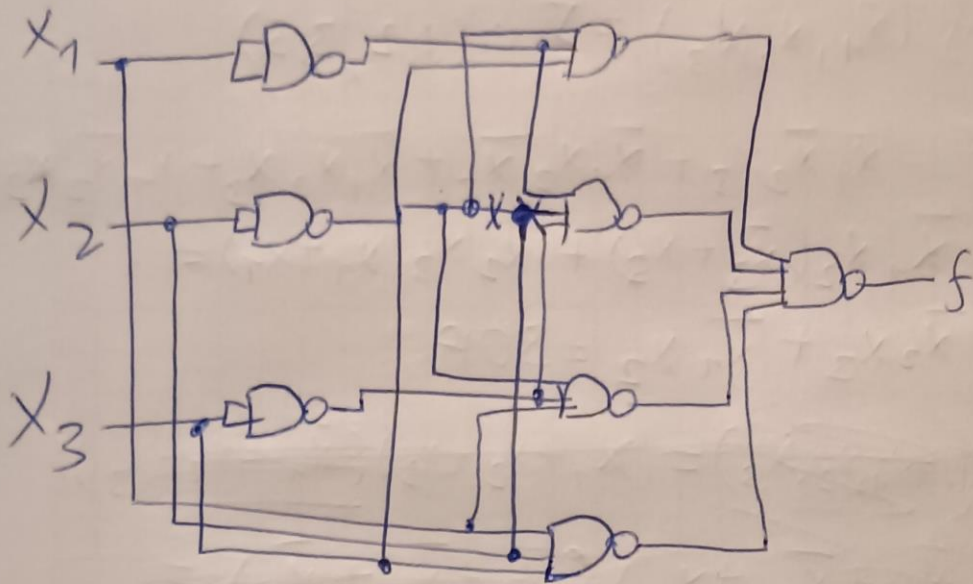
(2,3,1)

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

SOP:

$$f = \bar{x}_1 \bar{x}_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$$

(2,51)



(2, 3)

$$f(x_1, x_2, x_3) = \sum m(1, 2, 3, 5)$$

$$\begin{aligned} f &= \bar{x}_1 \bar{x}_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 = \\ &= \bar{x}_1 x_2 (\bar{x}_3 + x_3) + \bar{x}_2 x_3 (\bar{x}_1 + x_1) = \\ &= \bar{x}_2 x_3 + \bar{x}_1 x_2 = \text{SOP} \end{aligned}$$

~~$$f(x_1, x_2, x_3) = (x_1 + x_2 + \bar{x}_3)(\bar{x}_1 + x_2 + x_3)(\bar{x}_1 + \bar{x}_2 + x_3)$$~~

~~$$(x_1 + \bar{x}_2 + x_3)(\bar{x}_1 + \bar{x}_2 + \bar{x}_3) =$$~~

$\neq$

$$f(x_1, x_2, x_3) = (x_1 + x_2 + \bar{x}_3)(x_1 + \bar{x}_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)$$

$$(\bar{x}_1 + x_2 + \bar{x}_3) = (x_1 + x_2 + \bar{x}_3)(x_1 + \bar{x}_2 + x_3)$$

$$(x_1 + \bar{x}_2 + x_3)(\bar{x}_1 + x_2 + \bar{x}_3) =$$

$$= ((x_2 + \bar{x}_3) + x_1)((x_2 + \bar{x}_3) + \bar{x}_1)((x_1 + \bar{x}_2) + x_3)$$

$$((x_1 + \bar{x}_2) + x_3) = (x_2 + \bar{x}_3)(x_1 + \bar{x}_2)$$

POS